

Sooham Rafiz

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Career Break Independent Researcher and Study @ Fudan University *Jul 2025 - Present*

- Built and scaled [chat.runday.app](#), a multi-tenant agentic platform with heavy Kubernetes usage, currently supporting 50 small businesses with automated agentic workflows; designed for high availability and efficient resource allocation across workloads.
- Performance optimization projects on cache line investigation, CUDA, and kernel optimization. [Blog post on Apple M4 cache hierarchy](#)
- Replicated and extended (Noroozizdeh et al.) “Deep sequence models tend to memorize geometrically; it is unclear why.” [Blog post](#)
- Practical research into Pathstar optimization with mmap and sorting techniques. [Blog post](#)
- Took career break following significant investment gains, enabling focus on research interests and extended stay in Shanghai, China studying Chinese and Physics. Reached HSK level 5. 我在复旦学中文和物理, 都是因为好奇。达到 HSK 五级水平。
- Self-teaching: Quantum Physics, Classical Mechanics.

Short Story (YC'19) Senior Software Engineer / ML Engineer *Apr 2024 - Jul 2025*

- Leveraged open-source vision language model cogVLM-17B + LoRA with proprietary data to train, validate, test and deploy fine-tuned fashion cataloging and embedding utilities used throughout the company; automated cataloging of inventory. F1 score of 0.861 and accuracy of 88%.
- Designed and implemented a ShipmentEngine to track, bid and purchase 100k+ inbound and outbound shipments between company and vendors.
- Saved 1.8 million USD per annum in freight costs by switching away from USPS to Pitney Bowes E-commerce platform; led the full technology and engineering transition across Customers, Customer Support and Warehouse Operations.
- Designed an in-house cycle-counting system and supporting data ingestion pipelines to eliminate all inactive warehouse zones and improve inventory count accuracy.
- **Utilized:** Python, Pytorch, DDP, Postgres, Typescript, React, AWS, Falcon, Heroku

Amazon Software Development Engineer II - Payments Team *December 2021 - Nov 2023*

- Developed real-time Unpaid Time value algorithm and infrastructure for Germany — moving away from batched 3rd party vendor dependency. Speed up by a factor of 1000. Deployed to 2 million warehouse associates.
- Cross-functionally led the development of a timesheet analysis platform for 1k daily workers in Puerto Rico Amazon Logistic Center.
- Developed a new model of computation for missed meal payment, significantly reducing overpayment when warehouse associates miss meal breaks.
- Mentored engineers, authored team-specific onboarding documentation and video tutorials for future customers and new-hires.
- **Utilized:** Python, Java, AWS ECS, AWS DDB, Lambda, CloudWatch, Cloudformation, CDK

Qualcomm Machine Learning Acceleration Engineer *July 2019 - August 2021*

- Developed an end-to-end CI/CD ML test pipeline and performance dashboard to test model quantization on 100+ vision model families (Inceptionv3, MobileNet, YOLO etc.); profiled timing, memory footprint and quantization error across multiple hardware devices — enabling researchers to self-serve performance insights.
- Built a Python compiler to convert JSON computational graphs to standardized ONNX graphs supporting Qualcomm proprietary kernels for layer fusion and vector precision adjustment.
- Implemented a static quantization library to quantize floating-point neural network inputs and weights before inference for Snapdragon Neural Processor firmware.
- Reduced memory footprint of common operators such as Convolution and Argmax in compiled binaries.
- **Utilized:** C, Python, Pytorch, ONNX, Qualcomm Hexagon SDK, MLIR

Intel Corporation Deep Learning Acceleration Engineer (Intern) *Jun 2018 - Aug 2018*

- Reduced OpenVINO (a deep learning framework) compile time for CNN models by up to 24 hours using caching optimizations.

- Reduced compile time for deep convolutional model regression tests by caching intermediary C object files, libraries and configurations to a Python server, decreasing test time by 16 hours and improving repeatability.
- Migrated Intel's deep learning framework used by 50+ developers across backwards-incompatible environment updates, sustaining team productivity.
- **Utilized:** *C++, Python, OpenVINO, OpenCL*

Intel Corporation FPGA Development Engineer (Intern)

Jun 2017 - May 2018

- Created a comprehensive user interface and packaging system using Perl and Tcl, allowing parameter customization, code encryption and code generation; used in production by multiple customers.
- Built a C++ client and server to query FPGA registers via Direct Memory Access and Memory Mapped IO requests to FPGA workstations.
- Added FPGA hardware support to Intel's digital signal library by implementing C++ middleware.
- **Utilized:** *C++, Perl, Tcl, Quartus, Verilog*

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@skip Education

University of Toronto Honors B.Sc. Computer Science

2014 - 2019

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@skip Projects

[Hyperloglog Cardinality Estimator](#)

May 2019

- Compiled C++ HyperLogLog randomized algorithm used for efficient cardinality estimation.

[Image Inpainting C++ Library](#)

June 2016 - Aug 2016

- Authored the only open-source implementation of Criminisi et al.'s Inpainting algorithm in C++ as an online reference.

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@skip Technologies

Languages: Python, Java, Ruby, SQL, Typescript, Node.js

Infrastructure: Kubernetes (k3s, EKS), Docker, AWS (EC2, ECS, EKS, Lambda, DynamoDB, Cloudformation), Cloudflare, Infrastructure as Code

Tools & Other: Git, performance profiling & fine-tuning, React, PyTorch, ONNX, NoSQL